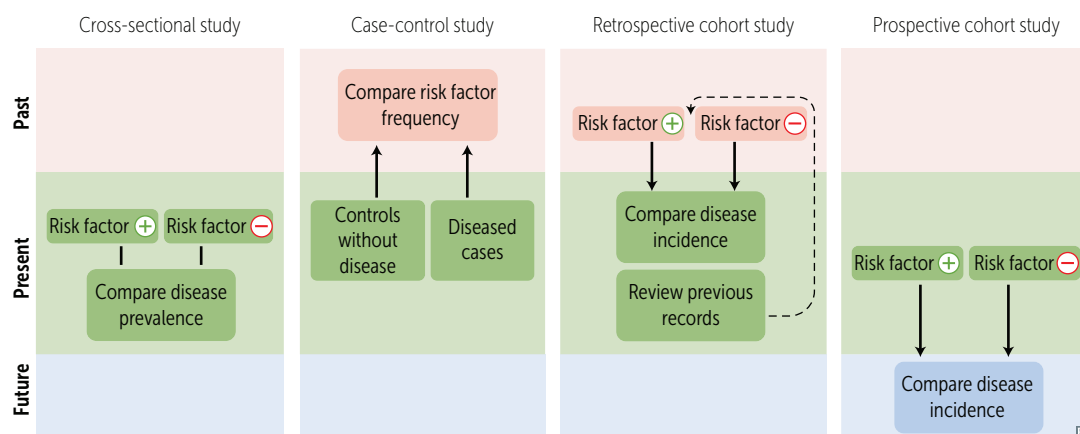


► PUBLIC HEALTH SCIENCES—EPIDEMIOLOGY AND BIOSTATISTICS

Observational studies

STUDY TYPE	DESIGN	MEASURES/EXAMPLE
Case series	Describes several individual patients with the same diagnosis, treatment, or outcome.	Description of clinical findings and symptoms. Has no comparison group, thus cannot show risk factor association with disease.
Cross-sectional study	Frequency of disease and frequency of risk-related factors are assessed in the present. Asks, “What is happening?”	Disease prevalence. Can show risk factor association with disease, but does not establish causality.
Case-control study	Retrospectively compares a group of people with disease to a group without disease. Looks to see if odds of prior exposure or risk factor differ by disease state. Asks, “What happened?”	Odds ratio (OR). Control the case in the OR . Patients with COPD had higher odds of a smoking history than those without COPD.
Cohort study	Compares a group with a given exposure or risk factor to a group without such exposure. Looks to see if exposure or risk factor is associated with later development of disease. Can be prospective or retrospective, but risk factor has to be present prior to disease development.	Disease incidence. Relative risk (RR). People who smoke had a higher risk of developing COPD than people who do not. Cohort = r elative r isk.
Twin concordance study	Compares the frequency with which both monozygotic twins vs both dizygotic twins develop the same disease.	Measures heritability and influence of environmental factors (“nature vs nurture”).
Adoption study	Compares behavioral traits/genetics in siblings raised by biological vs adoptive parents.	Measures heritability and influence of environmental factors.
Ecological study	Compares frequency of disease and frequency of risk-related factors across populations. Measures population data not necessarily applicable to individuals (ecological fallacy).	Used to monitor population health. COPD prevalence was higher in more polluted cities.



Clinical therapeutic trial

Experimental study involving humans. Compares therapeutic benefits of ≥ 2 interventions (eg, treatment vs placebo, treatment vs treatment). Study quality improves when clinical trial is randomized, controlled, and double-blinded (ie, neither subject nor researcher knows whether the subject is in the treatment or control group). Triple-blind refers to additional blinding of the researchers analyzing the data.

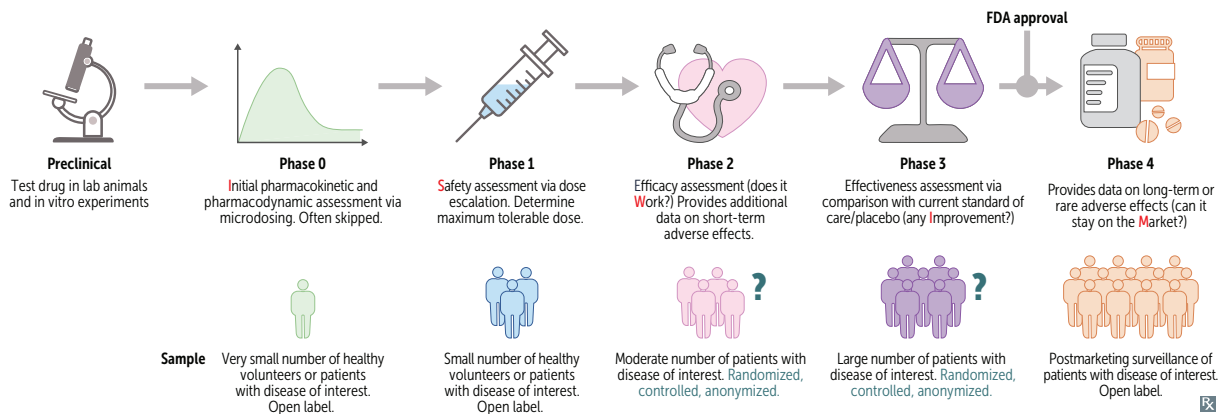
Crossover clinical trial—compares the effect of a series of ≥ 2 treatments on a subject. Order in which subjects receive treatments is randomized. Washout period occurs between treatments. Allows subjects to serve as their own controls.

Intention-to-treat analysis—all subjects are analyzed according to their original, randomly assigned treatment. No one is excluded, ie, once randomized, always analyzed. Attempts to avoid bias from attrition, crossover, and nonrandom noncompliance, but may dilute the true effects of intervention.

As-treated analysis—all subjects are analyzed according to the treatment they actually received. ↑ risk of bias.

Per-protocol analysis—subjects who fail to complete treatment as originally, randomly assigned are excluded. ↑ risk of bias.

Clinical trials occur after preclinical studies and consist of five phases (“Can I SWIM?”).



Off-label drug use

Use of a drug to treat a disease in a form, population group, or dosage that is not specifically approved by the FDA. Reasons for off-label use include treatment of an illness with no approved pharmacologic treatment or exploring alternative treatments after failure of approved options. Example: use of tricyclic antidepressants for treating neuropathic/chronic pain.

Bradford Hill criteria

A group of principles that provide limited support (ie, necessary but not sufficient criteria) for establishing evidence of a causal relationship between presumed cause and effect.

Strength	Association does not necessarily imply causation, but the stronger the association, the more evidence for causation.
Consistency	Repeated observations of the findings in multiple distinct samples.
Specificity	The more specific the presumed cause is to the effect, the stronger the evidence for causation.
Temporality	The presumed cause precedes the effect by an expected amount of time.
Biological gradient	Greater effect observed with greater exposure to the presumed cause (dose-response relationship).
Plausibility	A conceivable mechanism exists by which the cause may lead to the effect.
Coherence	The presumed cause and effect do not conflict with existing scientific consensus.
Experiment	Empirical evidence supporting the presumed cause and effect (eg, animal studies, in vitro studies).
Analogy	The presumed cause and effect are comparable to a similar, established cause and effect.

Quantifying risk

Definitions and formulas are based on the classic 2×2 or contingency table.

		Disease or outcome	
		⊕	⊖
Exposure or intervention	⊕	a	b
	⊖	c	d

TERM	DEFINITION	EXAMPLE	FORMULA								
Odds ratio	Typically used in case-control studies. Represents the odds of exposure among cases (a/c) vs odds of exposure among controls (b/d). OR = 1 → odds of exposure are equal in cases and controls. OR > 1 → odds of exposure are greater in cases. OR < 1 → odds of exposure are greater in controls.	If in a case -control study, 20/30 patients with lung cancer and 5/25 healthy individuals report smoking, the OR is 8; so the patients with lung cancer are 8 times more likely to have a history of smoking. You take a case to the OR .	$OR = \frac{a/c}{b/d} = \frac{ad}{bc}$ <table border="1"><tr><td>a</td><td>b</td></tr><tr><td>20</td><td>5</td></tr><tr><td>c</td><td>d</td></tr><tr><td>10</td><td>20</td></tr></table>	a	b	20	5	c	d	10	20
a	b										
20	5										
c	d										
10	20										
Relative risk	Typically used in cohort studies. Risk of developing disease in the exposed group divided by risk in the unexposed group. RR = 1 → no association between exposure and disease. RR > 1 → exposure associated with ↑ disease occurrence. RR < 1 → exposure associated with ↓ disease occurrence.	If 5/10 people exposed to radiation are diagnosed with cancer, and 1/10 people not exposed to radiation are diagnosed with cancer, the RR is 5; so people exposed to radiation have a 5 times greater risk of developing cancer. For rare diseases (low prevalence), OR approximates RR.	$RR = \frac{a/(a+b)}{c/(c+d)}$ <table border="1"><tr><td>a</td><td>b</td></tr><tr><td>5</td><td>5</td></tr><tr><td>c</td><td>d</td></tr><tr><td>1</td><td>9</td></tr></table>	a	b	5	5	c	d	1	9
a	b										
5	5										
c	d										
1	9										
Relative risk reduction	The proportion of risk reduction attributable to the intervention/treatment (ART) as compared to a control (ARC).	If 2% of patients who receive a flu shot develop the flu, while 8% of unvaccinated patients develop the flu, then RR = 2/8 = 0.25, and RRR = 0.75.	$RRR = 1 - RR$ $RRR = \frac{(ARC - ART)}{ARC}$								
Attributable risk	The difference in risk between exposed and unexposed groups.	If risk of lung cancer in people who smoke is 21% and risk in people who don't smoke is 1%, then the attributable risk is 20%.	$AR = \frac{a}{a+b} - \frac{c}{c+d}$ $AR\% = \frac{RR - 1}{RR} \times 100$								
Absolute risk reduction	The difference in risk (not the proportion) attributable to the intervention as compared to a control.	If 8% of people who receive a placebo vaccine develop the flu vs 2% of people who receive a flu vaccine, then ARR = 8%–2% = 6% = 0.06.	$ARR = \frac{c}{c+d} - \frac{a}{a+b}$								
Number needed to treat	Number of patients who need to be treated for 1 patient to benefit. Lower number = better treatment.		$NNT = 1/ARR$								
Number needed to harm	Number of patients who need to be exposed to a risk factor for 1 patient to be har med. Higher number = safer exposure.		$NNH = 1/AR$								
Case fatality rate	Percentage of deaths occurring among those with disease.	If 4 patients die among 10 cases of meningitis, case fatality rate is 40%.	$CFR\% = \frac{\text{deaths}}{\text{cases}} \times 100$								

Quantifying risk (continued)

TERM	DEFINITION	EXAMPLE	FORMULA
Mortality rate	Number of deaths (in general or due to specific cause) within a population over a defined period.	If 80 people in a town of 10,000 die over 2 years, mortality rate is 4 per 1000 per year.	Deaths/1000 people per year.
Attack rate	Proportion of exposed people who become ill.	If 80 people in a town are exposed and 60 people become ill, attack rate is 75%.	$\frac{\text{People who become ill}}{\text{Total people exposed}}$

Demographic transition	As a country proceeds to higher levels of development, birth and mortality rates decline to varying degrees, changing the age composition of the population.		
Population pyramid			
Birth rate	↑↑	↓	↓↓
Mortality rate	↑	↓	↓
Life expectancy	Short	Long	Long
Population	Growing	Stable	Declining

Likelihood ratio	$LR^+ = \frac{\text{probability of positive result in patient with disorder}}{\text{probability of positive result in patient without disorder}} = \frac{\text{sensitivity}}{1 - \text{specificity}} = \frac{\text{TP rate}}{\text{FP rate}}$
	$LR^- = \frac{\text{probability of negative result in patient with disorder}}{\text{probability of negative result in patient without disorder}} = \frac{1 - \text{sensitivity}}{\text{specificity}} = \frac{\text{FN rate}}{\text{TN rate}}$
	$LR^+ > 10$ indicates a highly specific test, while $LR^- < 0.1$ indicates a highly sensitive test.
	Pretest odds $\times$ LR = posttest odds. Posttest probability = posttest odds / (posttest odds + 1).

Kaplan-Meier curve	Used to estimate probability of survival over time. Graphic representation shows the survival probabilities (y-axis) vs length of time (x-axis). Useful for displaying “time-to-event” data. Outcomes examined may include any event, but frequently include mortality. Survival probability = 1 – (event probability). P value for the survival difference can be calculated using log rank test or Cox regression.	
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Evaluation of diagnostic tests

Sensitivity and specificity are fixed properties of a test. PPV and NPV vary depending on disease prevalence in population being tested.

Test efficiency =
 $(TP + TN) / (TP + FN + FP + TN)$

	Disease		
	(+)	(-)	
Test	(+)	TP	FP PPV = TP/(TP + FP)
	(-)	FN	TN NPV = TN/(TN + FN)
		Sensitivity = TP/(TP + FN)	Specificity = TN/(TN + FP)
		Prevalence TP + FN (TP + FN + FP + TN)	

Sensitivity (true-positive rate)

Proportion of all people with disease who test positive, or the ability of a test to correctly identify those with the disease.
 Value approaching 100% is desirable for **ruling out** disease and indicates a **low false-negative rate**.

$$= TP / (TP + FN)$$

$$= 1 - \text{FN rate}$$

SN-N-OUT = highly **Se**nsitive test, when **N**egative, rules **OUT** disease

High sensitivity test used for screening

Specificity (true-negative rate)

Proportion of all people without disease who test negative, or the ability of a test to correctly identify those without the disease.
 Value approaching 100% is desirable for **ruling in** disease and indicates a **low false-positive rate**.

$$= TN / (TN + FP)$$

$$= 1 - \text{FP rate}$$

SP-P-IN = highly **SP**ecific test, when **P**ositive, rules **IN** disease

High specificity test used for confirmation after a positive screening test

Positive predictive value

Probability that a person who has a positive test result actually has the disease.

$$PPV = TP / (TP + FP)$$

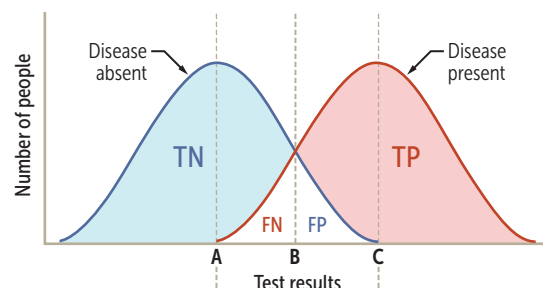
PPV varies directly with pretest probability (baseline risk, such as prevalence of disease):
 high pretest probability → high PPV

Negative predictive value

Probability that a person with a negative test result actually does not have the disease.

$$NPV = TN / (TN + FN)$$

NPV varies inversely with prevalence or pretest probability



Note: In diseases where diagnosis is based on lower values (eg, anemia), the TP and TN are switched in the graph, ie, ↓ sensitivity and ↓ NPV, and vice-versa.

Possible cutoff values for ⊕ vs ⊖ test result

- A = 100% sensitivity cutoff value
- B = practical compromise between specificity and sensitivity
- C = 100% specificity cutoff value

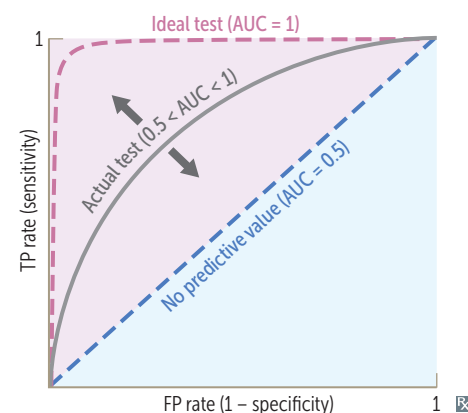
Lowering the cutoff value: ↑ Sensitivity ↑ NPV
 B → A (↑ FP ↓ FN) ↓ Specificity ↓ PPV

Raising the cutoff value: ↑ Specificity ↑ PPV
 B → C (↑ FN ↓ FP) ↓ Sensitivity ↓ NPV

Receiver operating characteristic curve

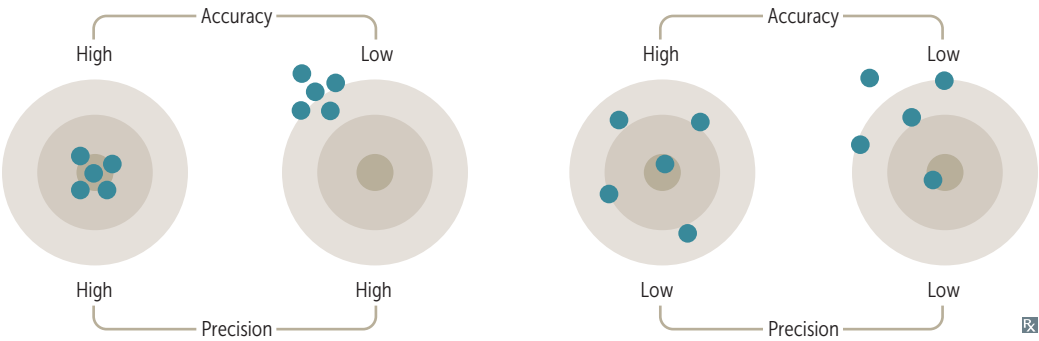
ROC curve demonstrates how well a diagnostic test can distinguish between 2 groups (eg, disease vs healthy). Plots the true-positive rate (sensitivity) against the false-positive rate ($1 - \text{specificity}$).

The better performing test will have a higher area under the curve (AUC), with the curve closer to the upper left corner.

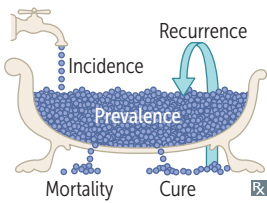


Precision vs accuracy

Precision (reliability)	The consistency and reproducibility of a test. The absence of random variation in a test.	Random error ↓ precision in a test. ↑ precision → ↓ standard deviation. ↑ precision → ↑ statistical power (1 – β).
Accuracy (validity)	The closeness of test results to the true values. The absence of systematic error or bias in a test.	Systematic error ↓ accuracy in a test.



Incidence vs prevalence



$\text{Incidence} = \frac{\# \text{ of new cases}}{\# \text{ of people at risk}} \quad (\text{per unit of time})$	Incidence looks at new cases (incidents).
$\text{Prevalence} = \frac{\# \text{ of existing cases}}{\text{Total \# of people in a population}} \quad (\text{at a point in time})$	Prevalence looks at all current cases.
$\frac{\text{Prevalence}}{1 - \text{prevalence}} = \text{Incidence rate} \times \text{average duration of disease}$	
Prevalence = incidence × duration of the disease.	Prevalence ~ pretest probability.
Prevalence > incidence for chronic diseases, due to large # of existing cases (eg, diabetes).	↑ prevalence → ↑ PPV and ↓ NPV.
	Prevalence ≈ incidence for short duration disease (eg, common cold).

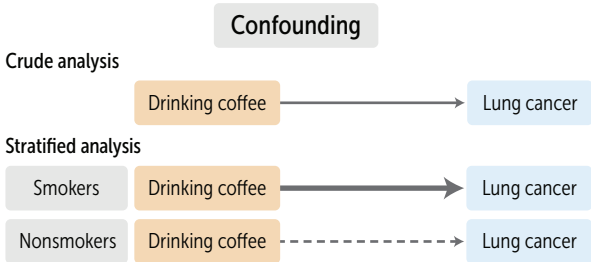
SITUATION	INCIDENCE	PREVALENCE
↑ survival time	—	↑
↑ mortality rate	—	↓
Faster recovery time	—	↓
Extensive vaccine administration	↓	↓
↓ risk factors	↓	↓
↑ diagnostic sensitivity	↑	↑
New effective treatment started	—	↓
↓ contact between infected and noninfected patients with airborne infectious disease	↓	↓

Bias and study errors

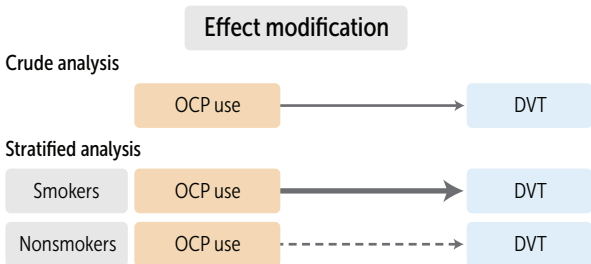
TYPE	DEFINITION	EXAMPLES	STRATEGIES TO REDUCE BIAS
Recruiting participants			
Selection bias	Nonrandom sampling or treatment allocation of subjects such that study population is not representative of target population Most commonly a sampling bias Convenience sampling—patients are enrolled on basis of ease of contact	Berkson bias—cases and/or controls selected from hospitals (bedside bias) are less healthy and have different exposures Attrition bias—participants lost to follow up have a different prognosis than those who complete the study	Randomization (creates groups with similar distributions of known and unknown variables) Ensure the choice of the right comparison/reference group
Performing study			
Recall bias	Awareness of disorder alters recall by subjects; common in retrospective studies	Patients with disease recall exposure after learning of similar cases	Decrease time from exposure to follow-up; use medical records as sources
Measurement bias	Information is gathered in a systemically distorted manner	Using a faulty automatic sphygmomanometer Hawthorne effect—participants change behavior upon awareness of being observed (Hawthorne watches you like a hawk).	Use objective, standardized, and previously tested methods of data collection that are planned ahead of time Use placebo group
Procedure bias	Subjects in different groups are not treated the same	Patients in treatment group spend more time in highly specialized hospital units	Blinding (masking) and use of placebo reduce influence of participants and researchers on procedures and interpretation of outcomes as neither are aware of group assignments
Observer-expectancy bias	Researcher’s belief in the efficacy of a treatment changes the outcome of that treatment (also called Pygmalion effect)	An observer expecting treatment group to show signs of recovery is more likely to document positive outcomes	
Interpreting results			
Lead-time bias	Early detection interpreted as ↑ survival, but the disease course has not changed	Breast cancer diagnosed early by mammography may appear to exaggerate survival time because patients are known to have the cancer for longer	Measure “back-end” survival (adjust survival according to the severity of disease at the time of diagnosis); a caveat of adjusting for severity is the potential masking of causality
Length-time bias	Screening test detects diseases with long latency period, while those with shorter latency period become symptomatic earlier	A slowly progressive cancer is more likely detected by a screening test than a rapidly progressive cancer	A randomized controlled trial assigning subjects to the screening program or to no screening

Confounding vs effect modification

TYPE	DEFINITION	EXAMPLES	STRATEGIES TO REDUCE BIAS
Confounding	Factor related to both exposure and outcome (not on a causal pathway) distorts the effect on the outcome No association after stratification	Coffee appears to be linked to lung cancer, but smoking is the true cause, as coffee drinkers tend to smoke more	Crossover studies (subject serves as self-control) Matching (patients with similar characteristics in both treatment and control groups) Analytic techniques (eg, regression analysis when confounding variables are known and were measured)
Effect modification	Exposure leads to different outcomes in subgroups stratified by factor True association exists	A study among women using OCPs showed significant risk of DVT, but when these data were stratified by smoking habits, there was a very strong association between OCP use and DVT among smokers, but there was no such association in people who do not smoke	Stratified analysis (eg, after testing for interaction between OCP and smoking, analyze risk among smokers and nonsmokers)



Note: Association disappeared after stratification.

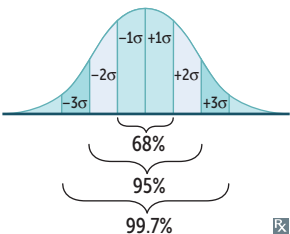


Note: Association was strong in one subgroup with weak/no association in the other subgroup.

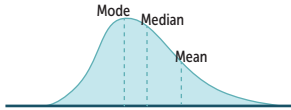
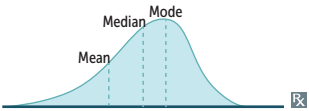
- Strong association
- Significant association
- - - - - Weak/no association



Statistical distribution

Measures of central tendency	Mean = (sum of values)/(total number of values).	Most affected by outliers (extreme values).
	Median = middle value of a list of data sorted from least to greatest.	If there is an even number of values, the median will be the average of the middle two values.
	Mode = most common value.	Least affected by outliers.
Measures of dispersion	Standard deviation = how much variability exists in a set of values, around the mean of these values.	σ = SD; n = sample size.
	Standard error = an estimate of how much variability exists in a (theoretical) set of sample means around the true population mean.	Variance = $(SD)^2$. $SE = \sigma/\sqrt{n}$. $SE \downarrow$ as $n \uparrow$.
Normal distribution	Gaussian, also called bell-shaped.	
	Mean = median = mode.	
	For normal distribution, mean is the best measure of central tendency.	
	For skewed data, median is a better measure of central tendency than mean.	

Nonnormal distributions

Bimodal distribution	Suggests two different populations (eg, metabolic polymorphism such as fast vs slow acetylators; age at onset of Hodgkin lymphoma; suicide rate by age).	
Positive skew	Typically, mean > median > mode. Asymmetry with longer tail on right; mean falls closer to tail.	
Negative skew	Typically, mean < median < mode. Asymmetry with longer tail on left; mean falls closer to tail.	

Statistical hypothesis testing

Null hypothesis	Also called H_0 . Hypothesis with no difference or association (eg, there is zero association between disease and risk factor in the population).
Alternative hypothesis	Also called H_1 . Hypothesis with at least one difference or relationship (eg, there is some association between disease and risk factor in the population).
P value	Probability of obtaining test results at least as extreme as those observed in the sample, assuming that H_0 is correct. Commonly accepted as 0.05 (< 5% of such repeated tests would show results that extreme just by chance alone).

Outcomes of statistical hypothesis testing

Correct result

Stating that there is an effect or difference when one exists (H_0 rejected in favor of H_1).

Stating that there is no effect or difference when none exists (H_0 not rejected).

	Reality	
	H_1	H_0
Study rejects H_0	Power (1 – β)	α Type I error
Study does not reject H_0	β Type II error	

Blue shading = correct result.

Testing errors

Type I error (α)

Stating that there is an effect or difference when none exists (H_0 incorrectly rejected in favor of H_1).

α is the probability of making a type I error (usually 0.05 is chosen). If $P < \alpha$, then assuming H_0 is true, the probability of obtaining the test results would be less than the probability of making a type I error. H_0 is therefore rejected as false.

Statistical significance $\neq$ clinical significance.

Also called false-positive error.

1st time boy cries wolf, the town believes there is a wolf, but there is not (false positive).

You can never “prove” H_1 , but you can reject the H_0 as being very unlikely.

Type II error (β)

Stating that there is not an effect or difference when one exists (H_0 is not rejected when it is in fact false).

β is the probability of making a type II error. β is related to statistical power (1 – β), which is the probability of rejecting H_0 when it is false.

↑ power and ↓ β by:

- ↑ sample size
- ↑ expected effect size
- ↑ precision of measurement
- ↑ α level (↑ statistical significance level).

Also called false-negative error.

2nd time boy cries wolf, the town believes there is no wolf, but there is one.

If you ↑ sample size, you ↑ power. There is **power in numbers**.

Generally, when type I error increases, type II error decreases.

Statistical vs clinical significance

Statistical significance—defined by the likelihood of study results being due to chance. If there is a high statistical significance, then there is a low probability that the results are due to chance.

Clinical significance—measure of effect on treatment outcomes. An intervention with high clinical significance is likely to have a large impact on patient outcomes/measures.

Some studies have a very high statistical significance, but the proposed intervention may have limited clinical impact/significance, eg, a study might show a statistical significance of lowered blood sugar levels by 1 mg/dL correlated with better outcomes, but this may not be clinically as important.

Confidence interval

Range of values within which the true mean of the population is expected to fall, with a specified probability.

CI = $1 - \alpha$. The 95% CI (corresponding to $\alpha = 0.05$) is often used. As sample size increases, CI narrows.

CI for sample mean = $\bar{x} \pm Z(SE)$

For the 95% CI, $Z = 1.96$.

For the 99% CI, $Z = 2.58$.

H_0 is rejected (and results are significant) when:

- 95% CI for mean difference excludes 0
- 95% CI OR or RR excludes 1
- CIs between two groups do not overlap

H_0 is not rejected (and results are not significant) when:

- 95% CI for mean difference includes 0
- 95% CI OR or RR includes 1
- CIs between two groups do overlap

Meta-analysis

A method of statistical analysis that pools summary data (eg, means, RRs) from multiple studies for a more precise estimate of the size of an effect. Also estimates heterogeneity of effect sizes between studies.

Improves power, strength of evidence, and generalizability (external validity) of study findings.

Limited by quality of individual studies and bias in study selection.

Common statistical tests**t-test**

Checks differences between **means** of **2** groups.

Tea is **mean**t for **2**.

Example: comparing the mean blood pressure between men and women.

ANOVA

Checks differences between means of **3** or more groups.

3 words: **AN**alysis **Of** **V**ariance.

Example: comparing the mean blood pressure between members of 3 different ethnic groups.

Chi-square (χ^2)

Checks differences between 2 or more percentages or proportions of **categorical** outcomes (not mean values).

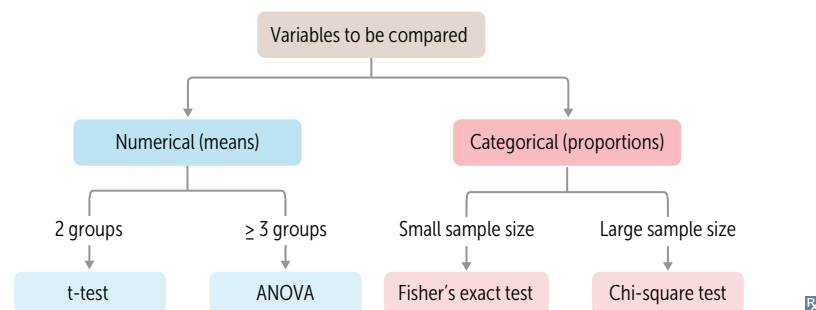
Pronounce **chi-tegorical**.

Example: comparing the proportion of members of 3 age groups who have essential hypertension.

Fisher's exact test

Checks differences between 2 percentages or proportions of categorical, nominal outcomes. Use instead of chi-square test with small samples.

Example: comparing the percentage of 20 men and 20 women with hypertension.



Pearson correlation coefficient

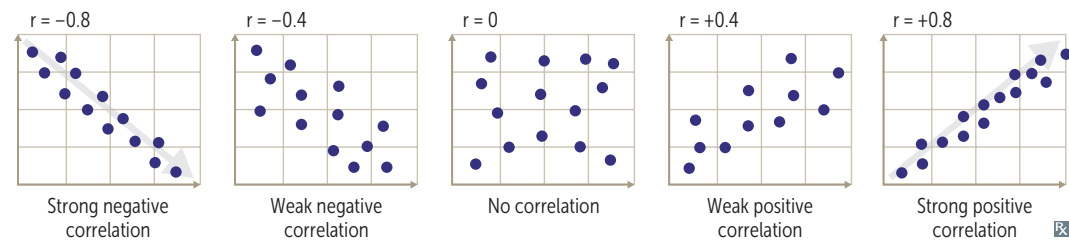
A measure of the linear correlation between two variables. r is always between -1 and $+1$. The closer the absolute value of r is to 1 , the stronger the linear correlation between the 2 variables.

Variance is how much the measured values differ from the average value in a data set.

Positive r value → positive correlation (as one variable ↑, the other variable ↑).

Negative r value → negative correlation (as one variable ↑, the other variable ↓).

Coefficient of determination = r^2 (proportion of variance in one variable that can be explained by variance in the other variable). Correlation does not necessarily imply causation.



► PUBLIC HEALTH SCIENCES—ETHICS

Core ethical principles

Autonomy	Obligation to respect patients as individuals (truth-telling, confidentiality), to create conditions necessary for autonomous choice (informed consent), and to honor their preference in accepting or not accepting medical care.
Beneficence	“Do good.” Physicians have a special ethical (fiduciary) duty to act in the patient’s best interest. May conflict with autonomy (an informed patient has the right to decide) or what is best for society (eg, mandatory TB treatment). Traditionally, patient interest supersedes. Principle of double effect—facilitating comfort is prioritized over potential side effects (eg, respiratory depression with opioid use) for patients receiving end-of-life care.
Nonmaleficence	“Do no harm.” Must be balanced against beneficence; if the benefits outweigh the risks, a patient may make an informed decision to proceed (most surgeries and medications fall into this category).
Justice	To treat persons fairly and equitably. This does not always imply equally (eg, triage).

Decision-making capacity

Physician must determine whether the patient is psychologically and legally capable of making a particular healthcare decision.

Note that decisions made with capacity cannot be revoked simply if the patient later loses capacity.

Intellectual disabilities and mental illnesses are not exclusion criteria unless the patient’s condition presently impairs their ability to make healthcare decisions.

Capacity is determined by a physician for a specific healthcare-related decision (eg, to refuse medical care).

Competency is determined by a judge and usually refers to more global categories of decision-making (eg, legally unable to make any healthcare-related decision).

Four major components of decision-making:

- Understanding (what do you know about your condition/proposed procedure/treatment?)
- Appreciation (what does your condition mean to you? why do you think your doctor is recommending this course of treatment?)
- Reasoning (how are you weighing your options?)
- Expressing a choice (what would you like to do?)